A preregistered short-run ladder fails to accelerate matched SHD learning

exp070 · 19 July 2026 · Draft

Abstract

This one-seed exploratory experiment asks whether any of three locked, ordered interventions makes both matched Dale-constrained COBA and PING networks learn SHD faster. Each candidate first passes a local 128/128 two-epoch smoke, then trains for five epochs on the same 7,340-utterance development-training partition and 816-utterance held-out validation set as exp069. The official test set remains sealed. No candidate clears the preregistered requirement of at least +3 percentage points and non-worse cross-entropy in both cells. The ladder therefore terminates without a forty-epoch run.

Methods

Matched comparison and locked ladder

Every cell uses seed 42, 256 excitatory and 64 inhibitory slots, fixed Dale-constrained recurrence, batch size 32, a membrane-mean readout scaled by 225, and identical input and readout dimensions. COBA disables the inhibitory loop and uses voltage-gradient dampening 1—the engine's no-dampening value. PING enables the loop and uses dampening 1000. The development split, sample order, checkpoint rule, raster positions, and all unnamed settings remain matched.

The candidates were evaluated in this preregistered order, stopping at the first promotion: (1) change only the timestep from 1 ms to 2 ms; (2) restore 1 ms and change only shared input-weight mean from 0.9 to 1.2; (3) restore input mean 0.9 and change only matched Adam learning rate from 0.0004 to 0.001. Candidates were never combined. Promotion required both cells at epoch five to gain at least three accuracy points over their own exp069 epoch-five result, with non-worse validation cross-entropy, finite training, no skipped update, and active non-saturated populations.

Results

Every candidate is killed

Candidate	Cell	Accuracy	Accuracy change	CE	CE change
2 ms	COBA	23.53%	−4.41 pp	2.678	0.206
2 ms	PING	27.7%	−4.9 pp	2.592	0.148
input mean 1.2	COBA	27.94%	0 pp	2.489	0.017
input mean 1.2	PING	33.7%	1.1 pp	2.446	0.002
learning rate 0.001	COBA	26.1%	−1.84 pp	2.486	0.014

Candidate	Cell	Accuracy	Accuracy change	CE	CE change
learning rate 0.001	PING	34.19%	1.59 pp	2.33	−0.115

Temporal coarsening harms both cells. Stronger input drive leaves COBA accuracy unchanged and improves PING by only 1.10 points, while both losses are slightly worse. The higher learning rate improves PING loss and adds 1.59 points, but COBA loses 1.84 points. Because the primary gate is joint, none is promoted.

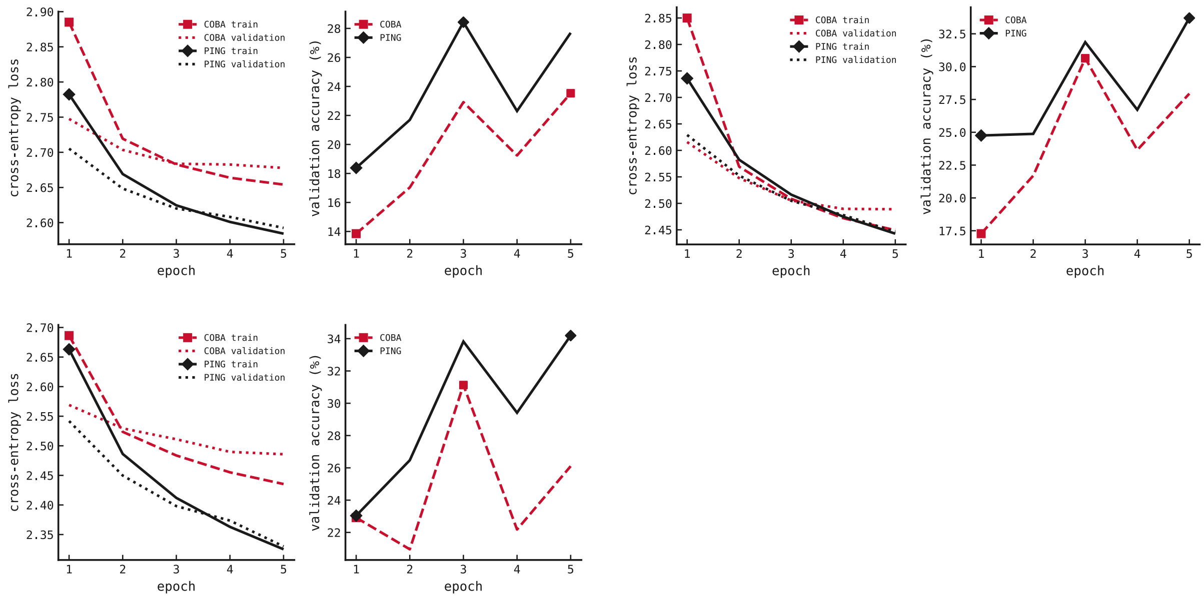

Figure 1: Five-epoch learning curves in registered order: 2 ms, input mean 1.2, and learning rate 0.001. The decision uses epoch five, not the visually best intermediate checkpoint.

Failures are scientific, not numerical

All six cells finish with finite losses and gradients, zero skipped updates, and zero non-finite forward batches. COBA excitatory rates span 14.22–30.64 Hz across the three epoch-five endpoints. PING remains active at 6.61–10.16 Hz E and 29.11–45.26 Hz I. These rates are far below the registered saturation thresholds. The failures therefore concern learning efficacy, not silence, saturation, or broken plumbing.

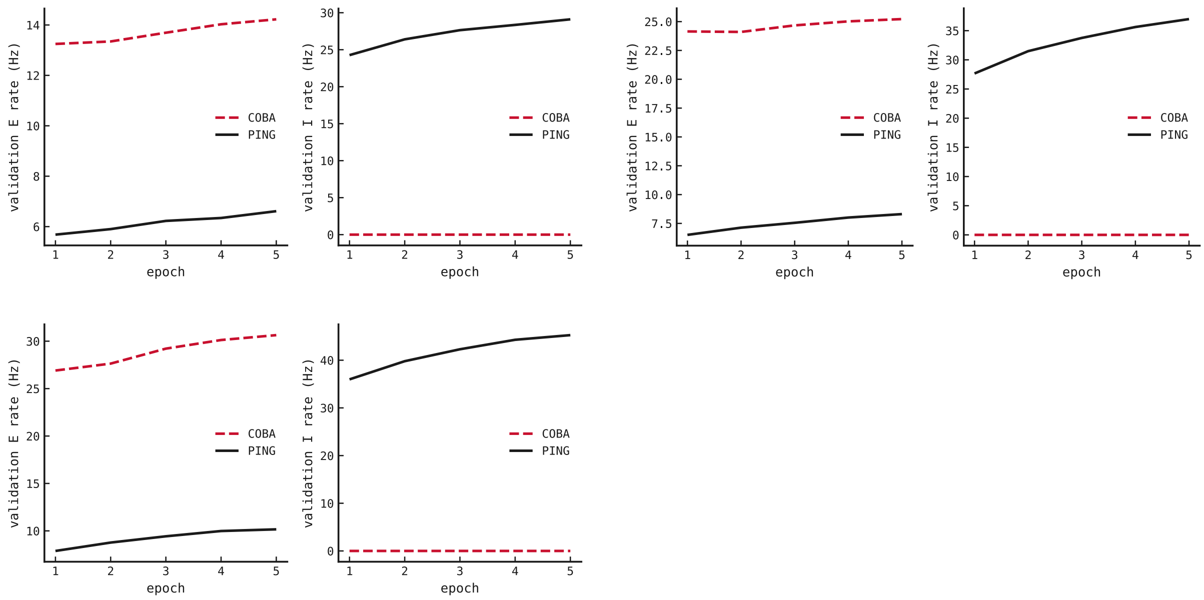

Figure 2: Held-out population firing rates for all three candidates. Stronger input and higher learning rate raise activity without approaching saturation.

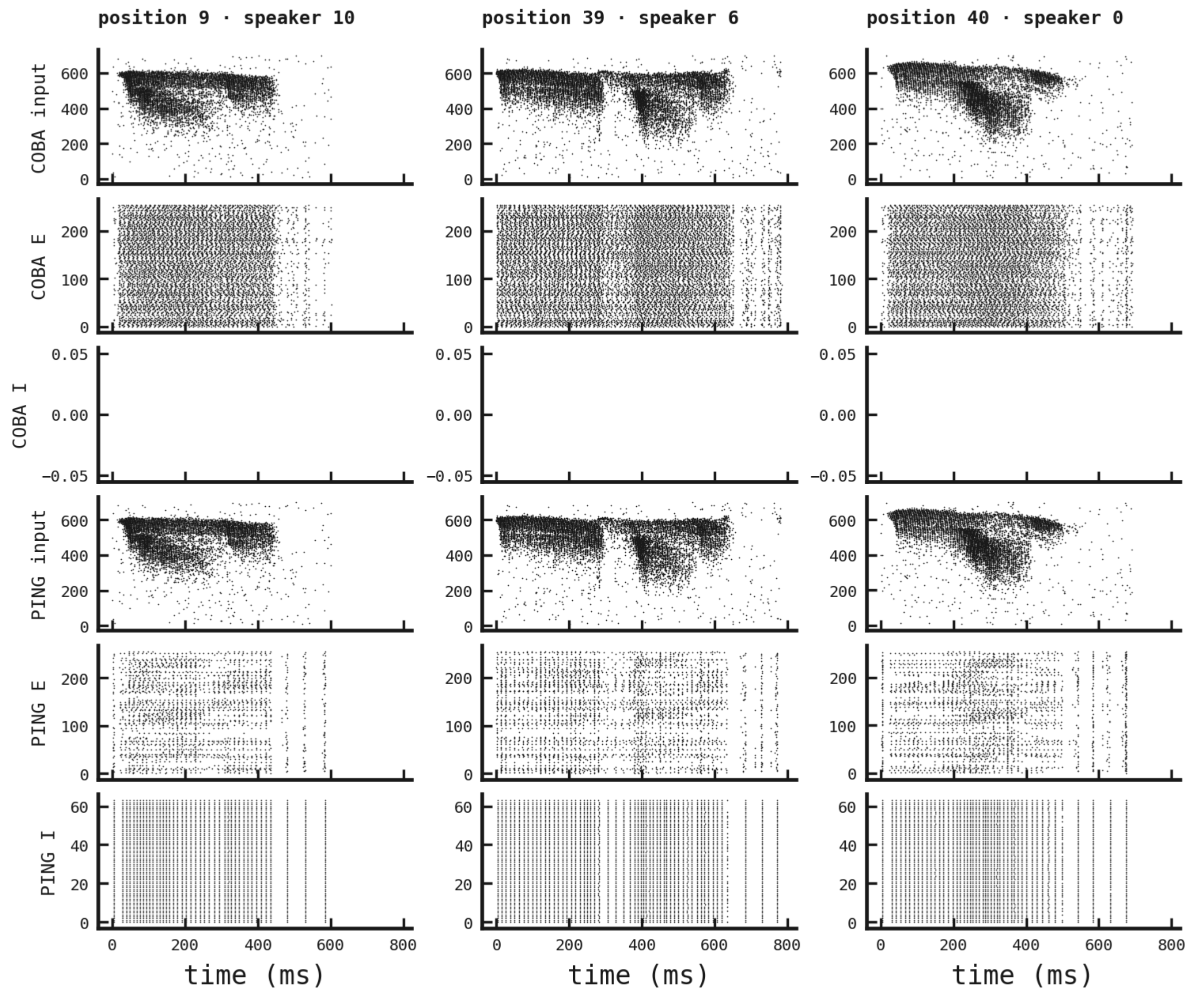

Figure 3: Matched validation rasters for the final candidate. Each COBA/PING column uses byte-identical binned input and fixed validation positions. The corresponding rasters for both earlier candidates are retained alongside their curves.

Verdict and limits

The preregistered exploratory hypothesis is not supported. None of the three isolated interventions improves both matched cells by at least three points at epoch five without worsening loss, so the protocol forbids the planned forty-epoch extension. The higher learning rate is the most promising PING signal, but its opposing COBA result means it is not a shared solution and cannot be promoted post hoc.

This is one seed and one held-out development split, deliberately appropriate for exploratory triage rather than a defended population claim. The official SHD test was neither staged nor loaded. Exact provider billing is recorded per attempt in the artifact bundle; final cumulative reconciliation remains pending provider settlement for the two later pairs.